MED 520 Examination
General Pathology and Disease Mechanisms

Instructions:

- Answer all questions.
- For Questions 1–4, choose the best option.
- For Questions 5–8, write brief but precise answers.

1. Which cellular adaptation involves an increase in cell size in response to increased functional demand?

(A) Hyperplasia

(B) Hypertrophy

(C) Metaplasia

(D) Dysplasia

2. Which type of necrosis is characteristically seen in tuberculosis infection?

(A) Liquefactive necrosis

(B) Coagulative necrosis

(C) Caseous necrosis

(D) Fat necrosis

3. In the inflammatory response, which cells are typically the first to arrive at the site of acute inflammation?

(A) Lymphocytes

(B) Macrophages

(C) Neutrophils

(D) Eosinophils

4. Which tumor marker is most commonly elevated in hepatocellular carcinoma?

(A) CA-125

(B) PSA (Prostate-Specific Antigen)

(C) AFP (Alpha-Fetoprotein)

(D) CEA (Carcinoembryonic Antigen)

5. Explain the difference between benign and malignant tumors in terms of growth patterns, invasion, metastasis, and differentiation. Why is metastatic potential the most critical distinguishing feature?

6. Describe the pathophysiology of atherosclerosis. What are the major risk factors, and how do they contribute to the formation of atherosclerotic plaques? Explain the role of endothelial dysfunction, lipid accumulation, and inflammation.

7. Discuss the cellular and molecular mechanisms of apoptosis (programmed cell death). How does it differ from necrosis, and what are the two major pathways that trigger apoptosis? Explain the role of caspases and the significance of apoptosis in development and disease.

8. Explain the concept of carcinogenesis and the multi-step model of cancer development. Describe the roles of oncogenes, tumor suppressor genes, and genes regulating apoptosis in the transformation from normal cells to malignant cells.